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Programming for Data Analytics 2

Part 2 – Classification and Model Improvement

Contents

Introduction

Machine learning plays a vital role in enabling organizations to make informed decisions based on data. One of the fundamental tasks in machine learning is classification, where a model assigns data points to predefined categories using input features.

This study addresses a binary classification problem within the healthcare domain, focusing on the classification of tumor samples as either *benign* or *malignant*. Since the dependent variable consists of discrete classes rather than continuous values, classification methods are most appropriate.

The main objective of this analysis is to construct a predictive model capable of accurately identifying tumor types based on their measured attributes. To achieve this, a systematic workflow is followed, including dataset selection, exploratory analysis, feature engineering, model development, and performance evaluation.

1. Dataset Selection and Justification

The dataset used in this study is the **Breast Cancer Wisconsin (Diagnostic) dataset**, sourced from Kaggle and originally obtained from the UCI Machine Learning Repository (UCI Machine Learning Repository, 1995).

This dataset was selected because it is highly suitable for classification tasks and has strong relevance to real-world healthcare applications. It consists of **569 instances and 30 numerical features**, all derived from digitized images of breast tissue samples. These features describe various characteristics of cell nuclei, including radius, texture, and concavity.

The target variable is binary, distinguishing between benign and malignant tumors, which aligns well with classification techniques such as Logistic Regression, Decision Trees, and Random Forest (Scikit-learn, 2024).

A key advantage of the dataset is its high usability, meaning it is clean, well-documented, and requires minimal preprocessing. Notably, it contains **no missing values**, allowing greater focus on analysis and modelling rather than data cleaning.

Additionally, this dataset is frequently used in academic research and machine learning studies, making it a reliable benchmark for comparison (GeeksforGeeks, 2025). Its application to a critical healthcare problem further enhances its practical significance.

Dataset Pitfalls and Mitigation

Although the dataset is of high quality, several limitations must be considered:

Table 1: Dataset Pitfalls and Mitigation Strategies

Issue	Explanation	Mitigation Approach
Multicollinearity	Certain variables (e.g. radius, area, perimeter) are strongly correlated	Apply correlation-based feature selection
Class imbalance	More benign than malignant cases	Use precision, recall, and confusion matrix instead of accuracy alone
Feature variation	Some variables show large ranges or skewed distributions	Use robust models (e.g. Random Forest) and explore via EDA

Despite these challenges, the dataset remains appropriate due to its structure, completeness, and relevance.

2. Algorithm Selection and Justification

Several classification algorithms were evaluated, including Logistic Regression, K-Nearest Neighbours (KNN), Decision Trees, and Random Forest. Among these, Random Forest was selected as the most suitable method.

Logistic Regression is easy to interpret but assumes linear relationships, which may not capture the complexity of this dataset (James et al., 2021). KNN is sensitive to noise and requires careful feature scaling (Han et al., 2011). While Decision Trees can model non-linear relationships, they are prone to overfitting.

Random Forest improves upon Decision Trees by combining multiple trees through ensemble learning, which enhances predictive performance and reduces overfitting (Breiman, 2001). It is well-suited for handling complex interactions and non-linear relationships commonly present in medical data.

Additional advantages include:

- Minimal need for feature scaling
- Robustness to noise
- Ability to rank feature importance

Although less interpretable than simpler models, its predictive strength makes it well-suited for this study.

3. Planning the Analysis

To ensure a thorough and effective modelling process, a structured analysis plan was designed. This framework outlines the key steps required to explore, prepare, and utilize the dataset for building a reliable classification model. The workflow follows a logical sequence, beginning with exploratory analysis and progressing through preprocessing, feature selection, model development, evaluation, and final reporting.

3.1 Exploratory Data Analysis (EDA)

The analysis begins with exploratory data analysis, which focuses on developing a comprehensive understanding of the dataset. This includes reviewing the number of records, identifying available features, and examining data types.

Descriptive statistics are generated to assess the behavior of each variable, including measures such as mean, spread, and variability. These insights help identify patterns and potential irregularities in the data.

In addition, visual techniques such as histograms, box plots, and correlation heatmaps are applied to uncover trends, highlight outliers, and explore relationships between variables. The distribution of the target variable (malignant versus benign) is also examined to determine whether any imbalance exists, as this can influence model performance.

3.2 Data Preprocessing

Before modelling, the dataset is prepared to ensure compatibility with machine learning algorithms. The independent variables (features) are separated from the dependent variable (target).

The data is then divided into training and testing subsets using an 80/20 split. This approach allows the model to learn patterns from one portion of the data while being evaluated on unseen data.

To ensure consistent and reproducible results, a fixed random seed is applied during the data splitting process.

3.3 Feature Selection

Feature selection is carried out to identify the variables that contribute most significantly to predicting the target outcome. This step improves model efficiency and reduces unnecessary complexity.

The process includes:

- Performing correlation analysis to identify relationships among variables

- Using feature importance scores from the Random Forest model
- Removing features that provide limited or redundant information

By refining the feature set, issues such as multicollinearity are minimized, and overall model performance is enhanced.

3.4 Model Training

The selected features are used to train a Random Forest classification model. This algorithm is applied to the training dataset to learn patterns and relationships between variables.

To improve predictive performance, hyperparameters are adjusted, including:

- The number of trees in the forest (*n_estimators*)
- The depth of each tree (*max_depth*)

Optimization techniques such as GridSearchCV can be used to determine the most effective parameter settings (Scikit-learn, 2024).

3.5 Model Evaluation

Once trained, the model is evaluated using the test dataset to determine how well it generalizes to new data. Several performance metrics are used to provide a balanced assessment, including:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Using multiple metrics is essential, particularly in cases where class imbalance exists, as accuracy alone may not fully reflect model effectiveness. Metrics such as recall are especially important in medical applications, where correctly identifying positive cases is critical (Han et al., 2011).

3.6 Model Improvement

To further refine the model, additional techniques are considered. These include:

- Hyperparameter tuning to optimize model settings
- Cross-validation methods, such as k-fold validation, to improve reliability

These approaches help ensure that the model performs consistently across different datasets and reduces the likelihood of overfitting.

3.7 Reporting

The final stage involves documenting the entire analysis process and presenting the results in a clear and structured manner. This includes summarizing findings from exploratory analysis, discussing model performance, and outlining any improvements made.

Visualizations and well-explained interpretations are used to effectively communicate results and support conclusions drawn from the analysis.

4 Conduct Analysis

4.1 Exploratory Data Analysis (EDA)

Dataset Overview

The dataset utilized in this study was obtained from Kaggle and is known as the Breast Cancer Wisconsin (Diagnostic) dataset, originally published by the UCI Machine Learning Repository (UCI Machine Learning Repository, 1995). Prior to analysis, the dataset was prepared by removing irrelevant attributes and transforming the target variable into a numerical format to ensure compatibility with machine learning algorithms.

The dataset contains a total of 569 records and 30 numerical features, each representing specific characteristics of tumor cell nuclei. In addition, a single target variable is included to indicate whether a tumor is benign or malignant. Since all variables are already in numerical form, no further encoding or transformation was required.

Preliminary statistical analysis revealed noticeable variation across the features, suggesting the presence of underlying patterns that can support classification tasks. Conducting exploratory data analysis at this stage is essential, as it provides insights into the dataset’s structure and behavior. This understanding helps guide subsequent steps, including feature selection and model development.

4.1.1 Alignment Between Planned and Conducted Analysis

Step from Question 3 (Planning)	**Description of Planned Task**	**What You Did in Question 4 (EDA)**
Structure and characteristics	Understand dataset size, features, and data types	Used df.info()
Summary statistics	Analyze central tendency, spread, and variability	Used df.describe()

Missing values check	Ensure dataset completeness and quality	Used df.isnull().sum()
Class distribution	Identify balance between benign and malignant cases	Used sns.countplot()
Feature distribution	Examine patterns and detect skewness in variables	Plotted histograms (feature distributions)
Correlation analysis	Identify relationships and multicollinearity	Used sns.heatmap()
Outlier detection	Detect extreme values that may affect the model	Used sns.boxplot()

4.1.2 Class Distribution

The target variable was examined to assess whether the dataset is balanced. The analysis indicates that there are more benign cases than malignant ones, which suggests a slight imbalance between the two classes.

Understanding the distribution of classes is an important step in classification problems, as an uneven distribution can influence the model to favor the majority class during prediction (Han *et al.*, 2011). To visualize this distribution, graphical tools from Python libraries such as Seaborn and Matplotlib were utilized.

4.1.3 Feature Distributions

The distribution of selected numerical features was analyzed using histograms. This method allows for a clear visual representation of how frequently different values occur within each variable and is commonly used in exploratory data analysis (IBM, 2023).

The results indicate that several features exhibit right-skewed distributions, where a large proportion of values are clustered toward lower ranges, with fewer observations at higher values. In contrast, some variables appear to follow a more symmetric, near-normal distribution, while others demonstrate noticeable skewness.

These differences highlight the diversity in statistical characteristics across the dataset, including variations in spread, central tendency, and skewness, which are important considerations for model development (James *et al.*, 2021).

4.1.4 Correlation Analysis

Relationships between variables were investigated using a correlation heatmap. The results reveal strong positive associations among features related to tumour size, including radius, perimeter, and area. The high correlation values suggest that these variables capture similar underlying information.

In addition to these strong relationships, moderate correlations were observed between variables such as compactness, concavity, and concave points, whereas some features showed weaker associations.

Identifying these relationships is essential, as highly correlated variables may introduce multicollinearity, which can negatively affect model performance and interpretability (James *et al.*, 2021).

4.1.5 Outlier Detection

Boxplots were used to detect outliers in selected features of the dataset. The results show the presence of several extreme values, particularly in the *area_mean* feature, where multiple observations extend significantly beyond the upper whisker of the plot.

This indicates that some tumor measurements are unusually large compared to the majority of the data. In addition, *area_mean* shows a much wider range compared to features such as *radius_mean* and *texture_mean*, highlighting differences in variability across features.

These outliers suggest variability in the dataset, which may influence model performance if not carefully considered.

4.1.6 Feature Distribution by Class

To further analyze how feature distributions differ between benign and malignant cases, boxplots were used to compare *area_mean* across the two classes.

The results show a clear separation between the groups, with malignant tumors generally having higher values than benign tumors. The median and overall distribution for malignant cases are significantly higher, indicating larger tumor size.

Additionally, malignant tumors exhibit greater variability and more extreme values, while benign cases are more tightly clustered.

This clear distinction between classes suggests that *area_mean* is a strong predictor of tumor classification and contributes to the model's effectiveness.

4.2 Feature Selection

Feature selection was undertaken to determine which variables have the strongest influence on predicting tumor diagnosis. This step is critical as it removes unnecessary or overlapping information, allowing the model to focus on the most informative inputs while improving efficiency and predictive performance (Scikit-learn, 2024; IBM, 2023).

To achieve this, correlation analysis was conducted to evaluate the strength of the relationship between each feature and the target variable. Variables showing strong associations with the diagnosis outcome were considered more relevant for the classification task. In particular, features such as *concave_points_worst*, *perimeter_worst*, *radius_worst*, and *area_worst* were identified as key predictors.

These variables are closely related to tumor size and structural characteristics, which are important indicators when distinguishing between benign and malignant cases. To refine the feature set further, a correlation threshold of **0.5** was applied. This ensured that only variables with moderate to strong relationships with the target variable were retained.

Applying this threshold helps to eliminate less informative features, reducing noise in the dataset and simplifying the modelling process. As a result, computational complexity is lowered, and the model is more likely to generalize effectively to unseen data (IBM, 2023; Scikit-learn, 2024).

Based on this filtering process, only features with correlation values exceeding the defined threshold were selected for model training, while those with weaker relationships were excluded from further analysis.

4.3 Model Training

The selected features were utilized to develop a Random Forest classification model. Initially, the dataset was divided into independent variables (X) and the dependent variable (y), where the selected attributes served as predictors and the diagnosis variable represented the outcome.

To evaluate the model effectively, the dataset was split into training and testing subsets using an 80/20 ratio. The training set consisted of 455 observations, while the remaining 114 observations were reserved for testing. This approach ensures that the model is trained on one portion of the data and validated on unseen data, allowing for an unbiased assessment of its performance.

A Random Forest classifier was then implemented and fitted to the training data. A fixed random state was applied during this process to guarantee consistency and reproducibility of results across multiple runs.

After training, the model was applied to the test dataset to generate predictions. These predictions were expressed as binary outputs, where a value of 0 represents a benign tumor and 1 represents a malignant tumor.

An example of the predicted output is shown below:

[0 1 1 0 0 1 1 1 1 0]

5. Evaluate Your Model

5.1 Interpretation and Evaluation of the Model

The performance of the model was assessed using the test dataset to determine how well it generalizes to new data. As outlined in the planning phase, multiple evaluation metrics were applied, including accuracy, precision, recall, F1-score, and the confusion matrix. These measures provide a comprehensive assessment of classification effectiveness, particularly in medical contexts where accurate detection is critical (Han *et al.*, 2011).

The model achieved an overall accuracy of **95.6%**, indicating that the majority of tumor cases were classified correctly. This demonstrates the effectiveness of the Random Forest algorithm in distinguishing between benign and malignant conditions.

The confusion matrix is presented below:

[[69 2]

 [3 40]]

This matrix provides a more detailed breakdown of the classification results:

- **69 benign cases** were correctly predicted as benign (True Negatives)
- **40 malignant cases** were correctly identified as malignant (True Positives)
- **2 benign cases** were incorrectly classified as malignant (False Positives)
- **3 malignant cases** were incorrectly classified as benign (False Negatives)

These results highlight that the model makes very few errors overall.

From a clinical perspective, false negatives are particularly critical because they represent cases where cancer is not detected. In this case, only three malignant cases were missed, indicating that the model performs well in identifying cancerous conditions.

In addition, the model achieved high precision (approximately 0.95–0.96), suggesting that most predicted labels are correct. The recall value for malignant cases is approximately 0.93, indicating that the model successfully detects the majority of cancer cases. The F1-score further confirms a strong balance between precision and recall.

Overall, the evaluation results demonstrate that the model provides reliable and accurate classifications, making it well-suited for this binary classification problem.

5.2 Retrain with Different Parameters (If Necessary)

Although the model demonstrates strong performance, it is important to evaluate whether further refinement through retraining is required. In this case, the model achieved an accuracy of **95.6%**, along with high precision, recall, and F1-score values. These results indicate that the model is already highly effective in distinguishing between benign and malignant tumor cases.

Given the limited number of misclassifications and the overall reliability of predictions, retraining the model is not considered essential at this stage. The current performance suggests that the model has successfully captured the underlying patterns within the dataset and can generalize well to unseen data.

However, there remains potential for further optimization. Improvements could be explored by adjusting key hyperparameters, such as the number of trees (*n_estimators*) and the maximum depth of each tree (*max_depth*). Fine-tuning these parameters may lead to incremental performance gains or enhance the model's ability to generalise across different datasets.

To systematically identify optimal parameter values, techniques such as **GridSearchCV** can be applied. This method evaluates multiple parameter combinations to determine the most effective configuration (Scikit-learn, 2024). In addition, **cross-validation approaches**, such as k-fold validation, can be used to ensure that the model produces consistent results across different data splits.

Overall, while retraining is not strictly required due to the model's already high performance, applying these optimization techniques could further strengthen model robustness and reliability.

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